A

Project Review Meeting

on

SMART HOSTEL & PG MANAGEMENT

By

B.MANASWINI	2420080029
Y.YUKTHA SAI	2420080060
SREE VARDHINI	2420030677
D. HEMESH	2420080037

Team Number: 16

Under the guidance of

Dr. P. Srilatha
Associate Professor

Department of Computer Science and Engineering

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Outline of the Presentation

- Abstract
- Introduction
- Problem Statement
- Project Flow Diagram
- Objectives with Justification
- Modules[Frontend web pages]
- Expected Outcomes of the Proposed Work
- Conclusion
- References

Abstract

The Smart Hostel & PG Management System is a web-based application designed to digitally manage hostel and PG activities in an efficient and user-friendly manner. The system replaces traditional manual processes such as registers and notice boards with a centralized online platform. It provides separate dashboards for administrators and students to manage rooms, rent status, complaints, and notices. The project aims to improve transparency, reduce paperwork, enhance communication, and ensure better management using modern web technologies like HTML, CSS, and JavaScript.

Introduction

Hostels and Paying Guest (PG) accommodations play a vital role in providing affordable housing to students and working professionals. However, most hostels still rely on manual record keeping, which leads to inefficiency, errors, and lack of transparency. Managing rooms, rent payments, complaints, and notices becomes difficult as the number of residents increases.

This project introduces a Smart Hostel & PG Management System that uses web technologies to simplify management tasks and improve the overall living experience.

Problem Statement

Traditional hostel and PG management systems face several challenges such as:

Manual maintenance of student records

Lack of proper complaint tracking

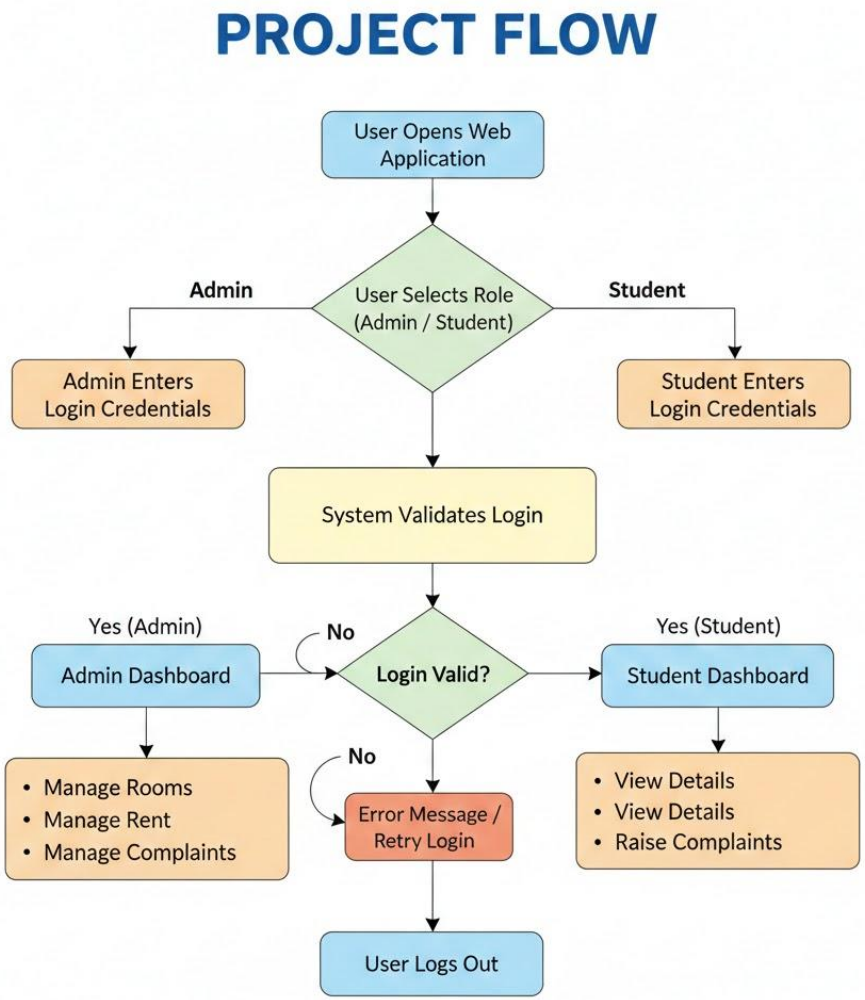
Inefficient rent payment monitoring

Poor communication between management and residents

Time-consuming and error-prone processes

There is a need for a smart, digital solution that automates these processes and provides easy access to information for both administrators and students.

Project Flow Diagram



Objectives with Justification

To digitize hostel/PG management

→ Reduces paperwork and manual effort

To provide role-based access (Admin & Student)

→ Ensures security and proper data access

To manage rooms and rent efficiently

→ Avoids confusion and payment delays

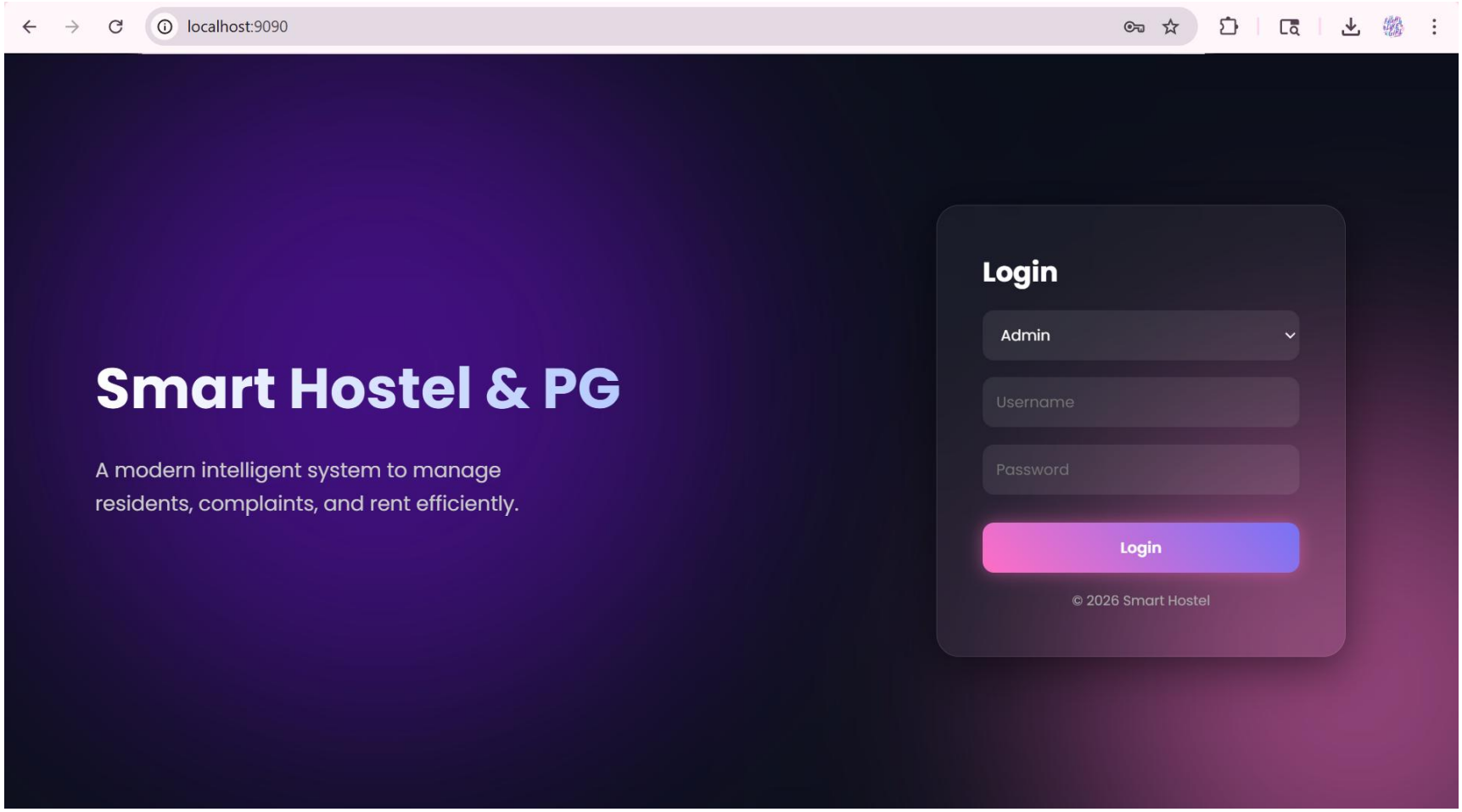
To implement a complaint management system

→ Ensures faster issue resolution

To improve communication using notices

→ Keeps students informed in real time

Modules [Frontend web pages]



localhost:9090/admin

Admin Dashboard

Add Resident

Manaswini

9347----

P-----

100

60000

Add

Residents

Load

Complaints

localhost:9090/admin

Residents

Load

- | - | Room - | ₹0
- | 9347668993 | Room 101 | ₹6000

Complaints

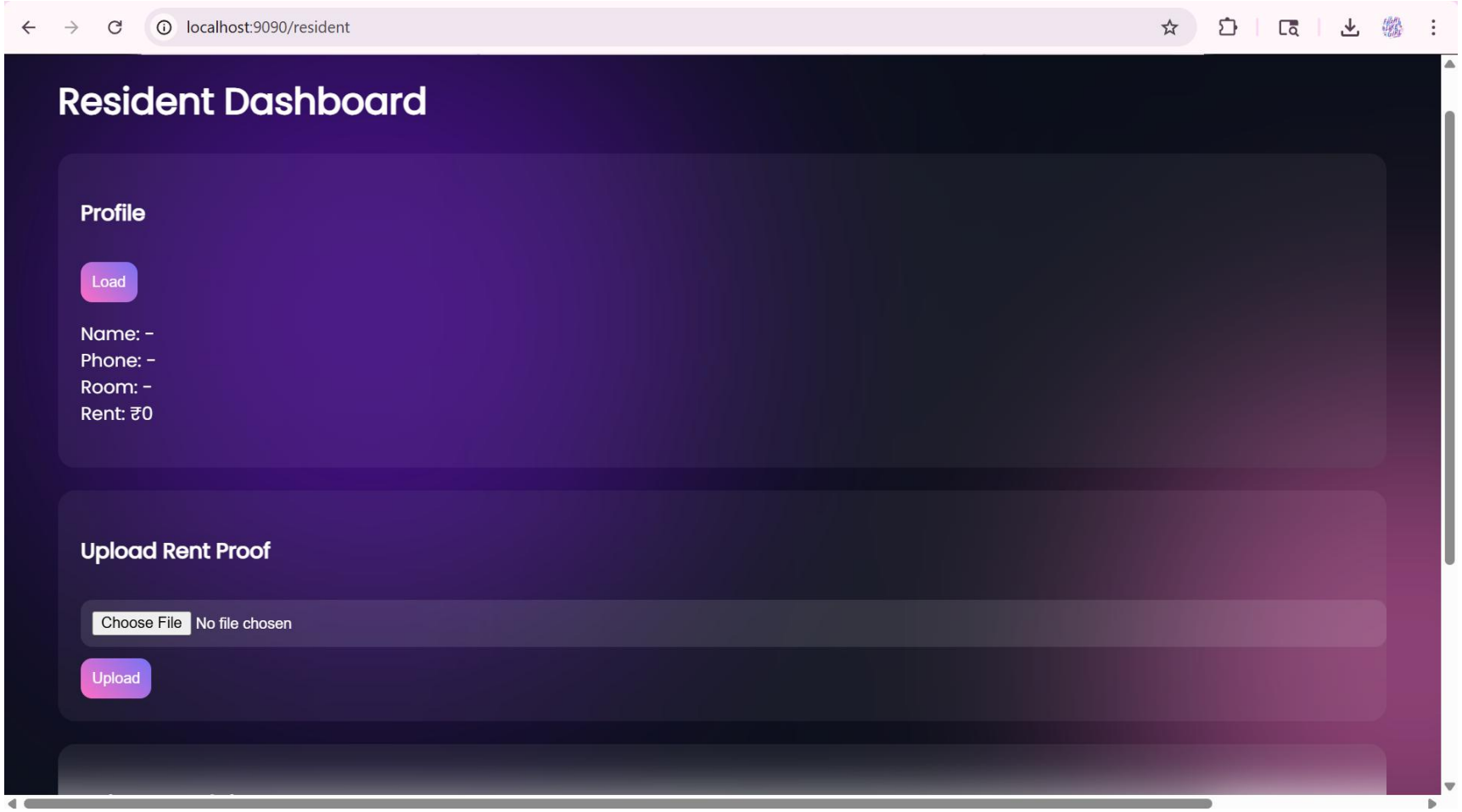
Load

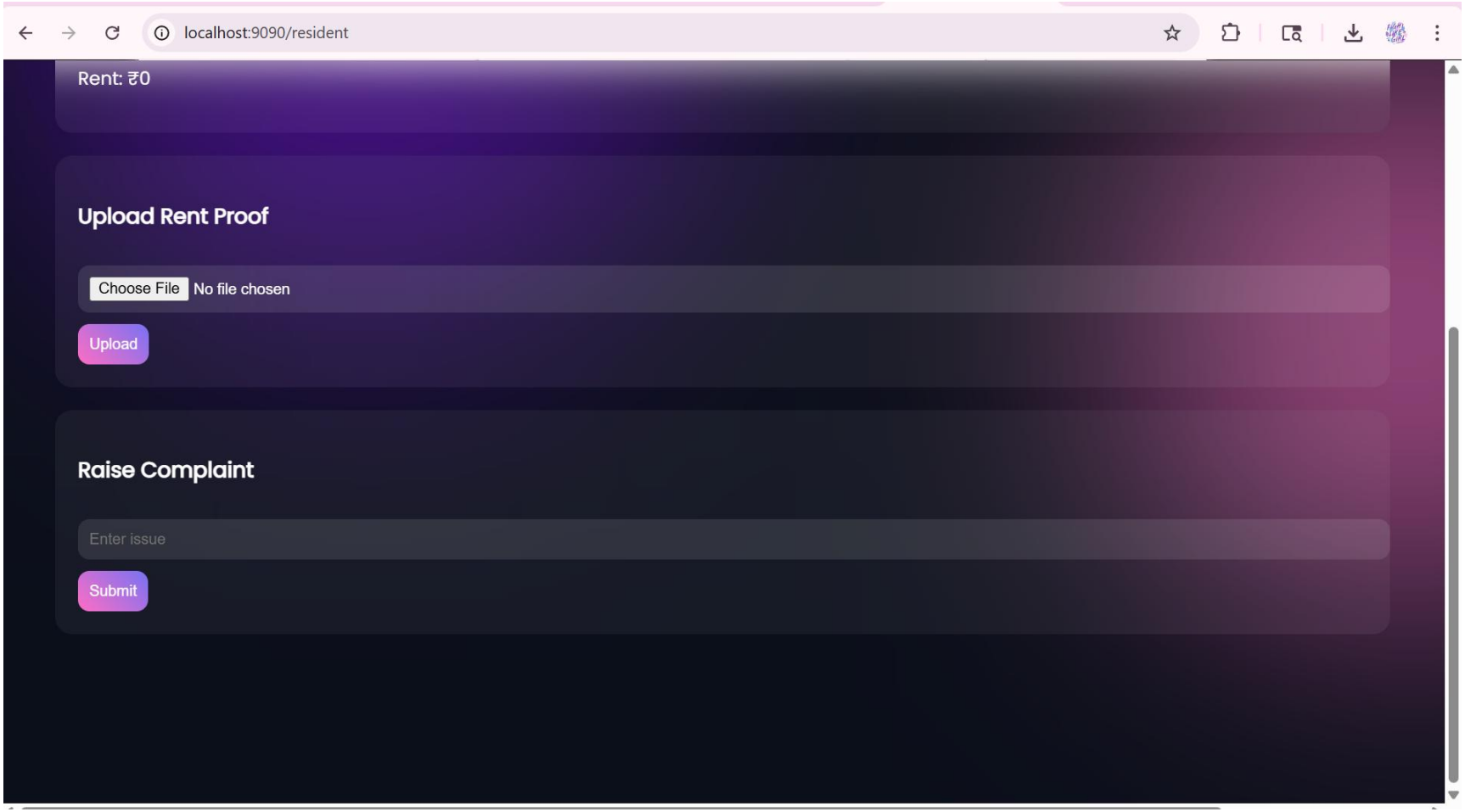
- : Water problem (Resolved)
Response: ok

Reply

Resident : WATER IS DIRTY!! (Resolved)
Response: solved

Reply


localhost:9090/resident
Resident Dashboard
Profile
Load
Name: -
Phone: -
Room: -
Rent: ₹0
Upload Rent Proof
Choose File
No file chosen
Upload


localhost:9090/resident
Rent: ₹0
Upload Rent Proof
Choose File
No file chosen
Upload
Raise Complaint
Enter issue
Submit

DATABASE:

MySQL 8.0 Command Line Cli

```
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 767
Server version: 8.0.44 MySQL Community Server - GPL

Copyright (c) 2000, 2025, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use hostel_db;
Database changed
mysql> show tables;
+---------------------+
| Tables_in_hostel_db |
+---------------------+
| complaint           |
| payment             |
| resident            |
| room                |
| student             |
+---------------------+
5 rows in set (0.01 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
5 rows in set (0.01 sec)

mysql> select * from complaint;
+----+------------------+--------------+---------------+----------+----------+
| id | issue            | student_name | resident_name | status   | response |
+----+------------------+--------------+---------------+----------+----------+
|  1 | Water problem    | Rahul        | NULL          | Resolved | ok       |
|  2 | WATER IS DIRTY!! | NULL         | Resident      | Resolved | solved   |
|  3 | FOOD NOT GOOD    | NULL         | Resident      | Pending  | NULL     |
|  4 |                  | NULL         | Resident      | Pending  | NULL     |
|  5 | NULL             | NULL         | NULL          | Pending  | NULL     |
|  6 | NULL             | NULL         | NULL          | Pending  | NULL     |
+----+------------------+--------------+---------------+----------+----------+
6 rows in set (0.01 sec)

mysql>  select * from payment;
+----+--------+--------------+
| id | amount | student_name |
+----+--------+--------------+
|  1 |   5000 | Rahul        |
+----+--------+--------------+
1 row in set (0.01 sec)

mysql>  select * from resident;
+----+---------+-------+------+------------+----------+-------------+--------------------+---------------+
| id | address | email | name | phone      | rent_due | room_number | payment_screenshot | payment_proof |
+----+---------+-------+------+------------+----------+-------------+--------------------+---------------+
|  1 | NULL    | NULL  | NULL | NULL       |        0 |           0 | NULL               | NULL          |
|  2 | p       | NULL  | NULL | 9347668993 |     6000 |         101 | NULL               | NULL          |
+----+---------+-------+------+------------+----------+-------------+--------------------+---------------+
```

MySQL 8.0 Command Line Cli

```
|  1 |   5000 | Rahul        |
+----+--------+--------------+
1 row in set (0.01 sec)

mysql>  select * from resident;
+----+---------+-------+------+------------+----------+-------------+--------------------+---------------+
| id | address | email | name | phone      | rent_due | room_number | payment_screenshot | payment_proof |
+----+---------+-------+------+------------+----------+-------------+--------------------+---------------+
|  1 | NULL    | NULL  | NULL | NULL       |        0 |           0 | NULL               | NULL          |
|  2 | p       | NULL  | NULL | 9347668993 |     6000 |         101 | NULL               | NULL          |
+----+---------+-------+------+------------+----------+-------------+--------------------+---------------+
2 rows in set (0.00 sec)

mysql>  select * from room;
+---------+----------------------+-----------+
| room_id | available            | room_type |
+---------+----------------------+-----------+
|     101 | 0x01                 | AC        |
+---------+----------------------+-----------+
1 row in set (0.00 sec)

mysql>  select * from student;
+----+-----------------+-------+------------+-------------+
| id | email           | name  | phone      | room_number |
+----+-----------------+-------+------------+-------------+
|  1 | rahul@gmail.com | Rahul | 9876543210 |         101 |
+----+-----------------+-------+------------+-------------+
1 row in set (0.00 sec)

mysql>
```

Expected Outcomes of the Proposed Work

1 Improved efficiency in hostel/PG management

2 Reduced manual errors and paperwork

3 Faster complaint handling

4 Better transparency in rent management

5 User-friendly interface for students and admins

6 Scalable system for future backend integration

Conclusion and Future Scope

The Smart Hostel & PG Management System provides an effective solution to the problems faced in traditional hostel management. By using modern web technologies, the system ensures efficient data handling, improved communication, and better user experience. This project demonstrates how simple frontend technologies can be used to design a smart and innovative management system. The application can be further enhanced by integrating backend services and databases for real-world deployment.

References

W3Schools

Reference for HTML, CSS, and JavaScript basics

MDN Web Docs

Reference for JavaScript functions, DOM manipulation, and events

GeeksforGeeks

Reference for web project ideas and frontend logic

freeCodeCamp

Reference for frontend design concepts and best practices

Stack Overflow

Reference for solving JavaScript and UI-related doubts

Thank you